Final Report

ISBA 2403 - Machine Learning With Python

Bronco Health Analytics

Screen Time, Sleep & Stress Analysis Data Case Study

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1. Problem Statement & Motivation

Advancements in digital technology have significantly increased the amount of time individuals spend on smartphones and other digital devices. While these technologies have improved connectivity and productivity, excessive screen time has been increasingly associated with adverse mental health outcomes, particularly elevated stress levels. This growing concern positions stress as a critical public health issue in the modern digital era.

According to the American Psychological Association (2007), approximately 77% of individuals regularly experience physical symptoms caused by stress, while 33% report experiencing extreme levels of stress. Furthermore, workplace stress alone is estimated to cost the United States nearly $300 billion annually, as reported by the American Institute of Stress (2022). These statistics highlight the widespread and significant impact of stress on both individuals and organizations.

Given this context, understanding the relationship between digital behavior such as screen time, sleep patterns, and lifestyle habits and stress levels has become an important area of study within behavioral analytics and digital health research.

This project aims to address two key objectives:

1. **Predictive Objective:**
   To determine whether behavioral factors such as screen time, sleep duration, and related variables can effectively predict an individual's stress level using machine learning techniques.
2. **Methodological Objective:**
   To evaluate how machine learning models trained on synthetic data perform when applied to real-world data collected through surveys. This includes assessing the generalizability and reliability of models across different data distributions.

The machine learning task is formulated as a **binary classification problem**, where the target variable perceived stress level is initially measured on a scale from 0 to 10 and subsequently categorized into two classes: **low stress** and **high stress**.

This problem is particularly relevant because it addresses a critical challenge in modern machine learning applications: the reliance on synthetic datasets. While synthetic data offers advantages such as scalability and accessibility, its effectiveness depends on how well it represents real-world patterns. Evaluating model performance across synthetic and real-world data is therefore essential to understanding the practical applicability of such systems.

Ultimately, this study contributes to both **applied machine learning** and **digital health research** by exploring the reliability of synthetic data while identifying key behavioral indicators of stress.

2. Data Collection

To evaluate both predictive performance and real world applicability, this study utilizes a **dual data approach** consisting of a synthetic dataset and a secondary survey dataset.

2.1 Synthetic Dataset (Kaggle)

A publicly available dataset from Kaggle was used as the primary training source.

- Contains behavioral and lifestyle features such as:
  - Screen time
  - Sleep duration
  - Work/study hours
  - Physical activity
- Includes a stress level score (0 to10 scale)

Limitation: The dataset is synthetic in nature, meaning it may not fully capture real-world behavioral patterns, biases, or noise. This raises concerns about the external validity of models trained solely on this data.

2.2 Real World Dataset (Class Survey)

To complement the synthetic dataset, a survey was conducted to collect real world data on behavioral patterns and perceived stress levels.

- Participants: 61 students (classroom-based sample)
- Data Collection Method: Structured questionnaire

To ensure consistency, the survey was designed to closely align with the feature set of the Kaggle dataset, enabling direct comparison and model evaluation.

Limitation: The dataset is subject to limitations including a small sample size, potential self-reporting bias, and inherent variability in responses.

2.3 Overview

	Kaggle Dataset	**Real World Data**
Number of Records	5100	61
Number of Features	16	11

3. Data Cleaning

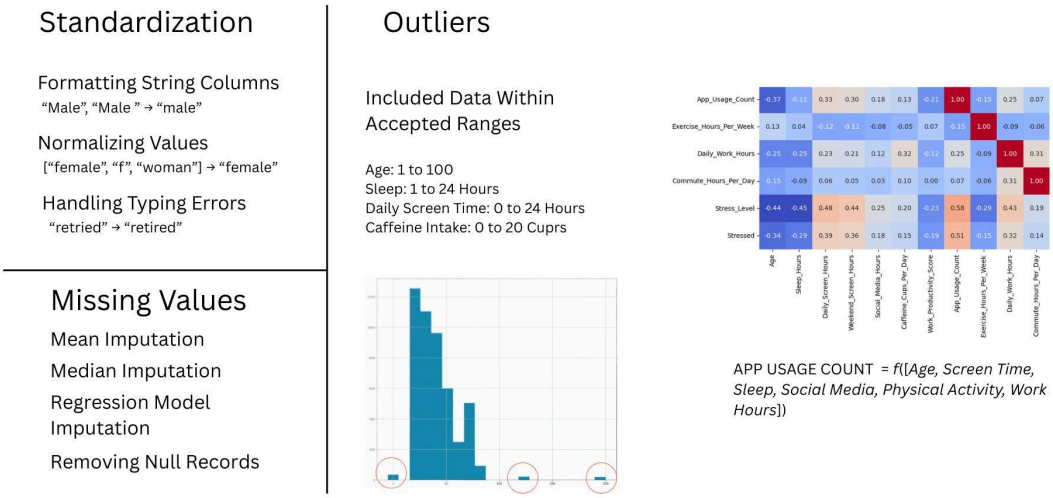

Figure 1. Data Cleaning Process

To ensure data quality and consistency, a structured data preparation pipeline was applied, focusing on three key aspects: standardization, outlier handling, and missing value treatment (as shown in Figure 1).

Categorical inconsistencies such as formatting differences, value variations, and typographical errors were standardized to ensure uniform representation across features. Numerical variables were validated against logical ranges, and extreme or invalid observations were removed to maintain realistic behavioral patterns.

Missing values were addressed using a combination of statistical and model-based techniques. While common features were imputed using central tendency measures, more complex dependencies such as app usage behavior were estimated using a regression based approach.

Imputation Technique	Target Feature	Justification
Mean Imputation	• Social Media Hours • Exercise Hours • Work Hours • Commute Time	Follows an approximate normal distribution and thus can be imputed using mean.
Median Imputation	• Productivity Score	To preserve the distribution and central tendency work.
Regression Based Imputation	• App Usage Count	Since this feature has significant correlation with other features, a regression model is trained on those features to predict the app usage count.

This pipeline ensured that both synthetic and real world datasets were transformed into a clean, consistent, and analysis ready format. After cleaning, the final dataset consisted of:

	Kaggle Dataset	**Real World Data**
Number of Records	3957	56
Number of Features	16	11

4. Exploratory Data Analysis

Exploratory analysis was conducted to understand the distribution of stress levels, identify key behavioral patterns, and compare trends between the synthetic (Kaggle) and real-world (survey) datasets.

4.1 Distribution of Stress and Behavioral Variables

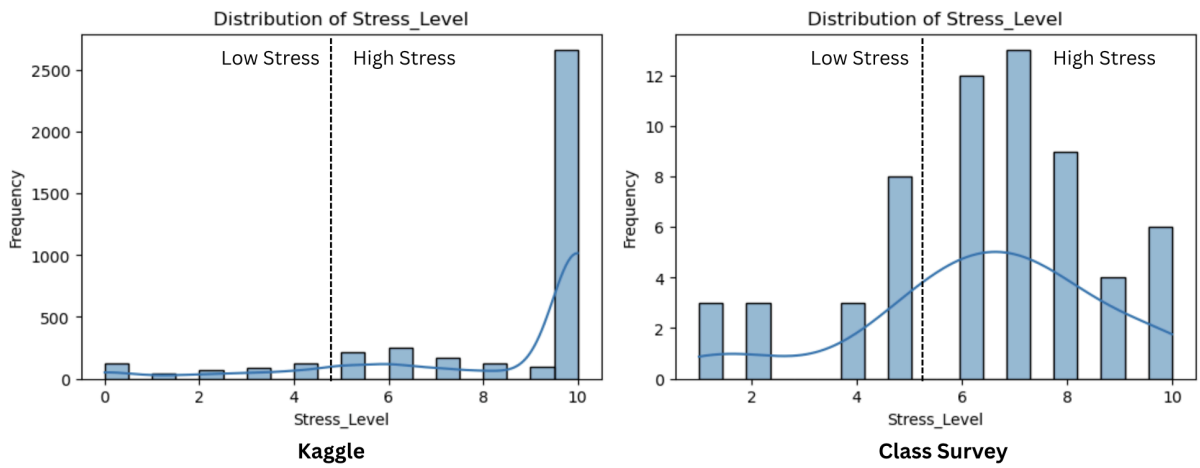

Figure 2. Stress Level Distribution

The distribution of stress levels in the Kaggle dataset is highly skewed, with a strong concentration at the maximum value (10), indicating a dominance of high stress observations. In contrast, the survey data exhibits a more gradual distribution, with stress levels spread across the 5 to 10 range, reflecting more realistic variability in perceived stress. Despite the differences observed in stress level distribution, most independent variables exhibited similar distribution patterns across both datasets, suggesting that the synthetic data captures general behavioral trends reasonably well.However, the variation in the target variable highlights a potential gap in how stress is represented in synthetic data compared to real-world observations.

4.2 Screen Time vs Stress Level Scatter Plot

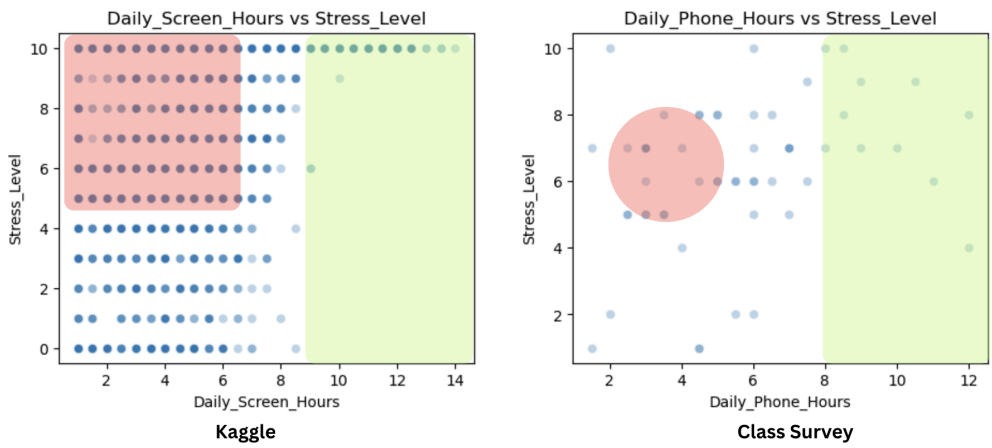

Figure 3. Screen Time vs Stress Level

The relationship between screen time and stress levels shows a clear distinction between synthetic and real-world data patterns. In the Kaggle dataset, stress levels increase sharply beyond a screen time threshold (approximately 8 to 9 hours), with a dense concentration of high stress values at higher usage levels. This indicates a strong and structured positive relationship between screen time and stress. In contrast, the survey data exhibits a more dispersed pattern. While higher screen time (8+ hours) is generally associated with elevated stress, the relationship is less rigid, with significant variability across individuals. Moderate screen time (3–6 hours) also shows a wide spread of stress levels, reflecting real-world behavioral diversity.

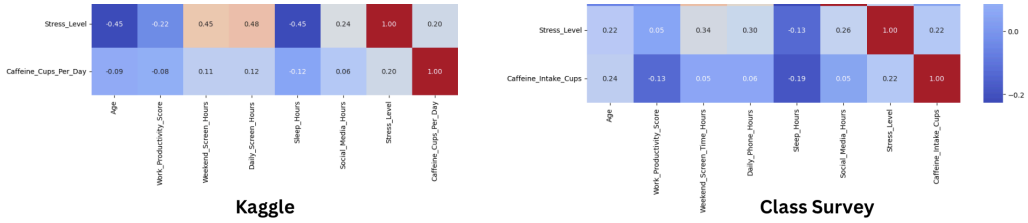

Figure 4. Correlation Map

Correlation analysis revealed consistent behavioral patterns across both datasets. Screen-related variables exhibited a positive association with stress levels, while sleep duration showed a negative association. In the Kaggle dataset, **daily screen hours, and weekend screen hours** demonstrated the strongest positive correlations with stress, whereas **sleep hours** showed the strongest negative relationship. Similarly, in the survey data, **weekend screen time, daily phone usage, and social media hours** were most strongly associated with higher stress levels.

5. Model Development

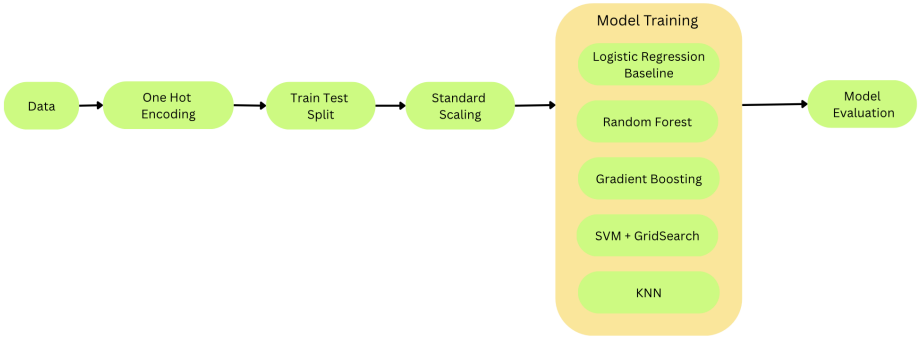

Figure 5. Machine Learning Framework

Several machine learning models were implemented to evaluate their effectiveness in predicting stress levels based on behavioral and demographic features. The models considered include **Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**.

Prior to training, categorical variables were transformed using **one-hot encoding** to ensure compatibility with machine learning algorithms, while numerical features were standardized using **feature scaling** to maintain consistent value ranges across variables.

The dataset was structured into predictor variables and a target variable, where features such as screen time, sleep duration, social media usage, caffeine intake, and demographic attributes were used to predict categorized stress levels (low vs high). An 80/20 train-test split was applied to evaluate model performance.

Logistic Regression was used as a baseline model to establish a reference for performance. Tree-based models such as Random Forest and Gradient Boosting were employed to capture non-linear relationships and feature interactions, particularly combinations of behavioral factors influencing stress. SVM was used to identify an optimal decision boundary separating stress categories in the feature space, while KNN leveraged similarity between individuals to make predictions based on neighboring observations.

This diverse set of models enables comparison across linear, distance-based, and ensemble approaches, providing a comprehensive evaluation of predictive performance.

6. Results

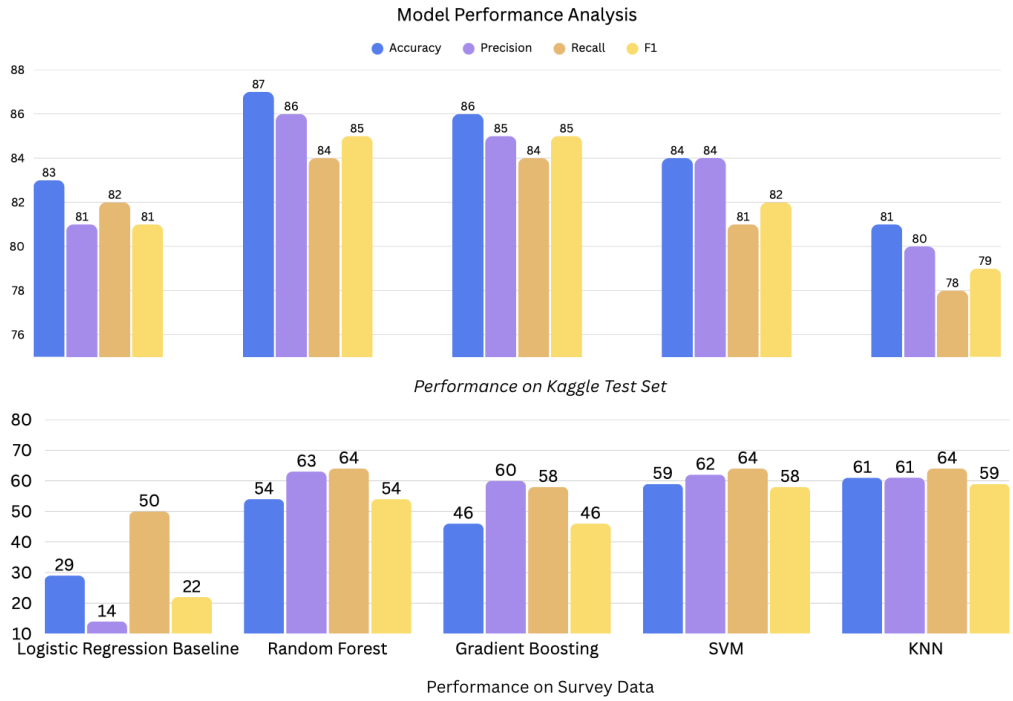

Figure 6. Model Performance on Kaggle Test Data (top) & Survey Data (below)

The comparative evaluation of five predictive models on the Kaggle holdout test set demonstrates a clear advantage for tree-based ensemble methods over linear and distance-based classifiers. **Random Forest** emerged as the top-performing model, providing a strong balance between Precision and Recall, making it the most reliable candidate for real-world deployment.

Key Insights:

- **Ensemble Superiority:** Both Random Forest and Gradient Boosting delivered the highest predictive performance. Random Forest slightly outperformed Gradient Boosting with an **Accuracy of 87%**, **Precision of 86%**, and **F1 score of 85%**, highlighting its robustness in minimizing both false positives and false negatives.
- **Baseline Benchmark:** Logistic Regression provided a solid foundation (**Accuracy: 83%, F1: 81%**), but the ensemble models' 3 to 4% improvement justifies their additional computational complexity for this dataset.
- **Underperformers:** KNN lagged behind all other models, especially in Recall (**78%**), suggesting challenges with the dataset's dimensionality. SVM showed modest improvements over the baseline (**Accuracy: 84%**) but still fell short of tree-based models.

- **Strategic Metric Alignment:** In scenarios where missing high-stress instances is costly, Recall is critical. Both Random Forest and Gradient Boosting achieved the highest Recall (**84%**), but Random Forest's higher Precision (**86%**) reduces false alarms, supporting its selection for deployment.

To validate each model's practical applicability, **it was next evaluated on real-world behavioral and demographic data** collected from users. This step ensures that the model's strong performance on the Kaggle dataset translates effectively to live scenarios, capturing genuine stress patterns while maintaining balanced error rates. The results show a substantial decline in performance across all models compared to their performance on the Kaggle test set. Among the models, **K-Nearest Neighbors (KNN)** achieved the highest accuracy at approximately **61%**. It correctly identified **64% of high-stress instances (Recall)** and achieved a strong **Precision of 61%**, indicating reliable predictions when classifying high-stress cases.

Support Vector Machine (SVM) performed similarly, with an **accuracy of 59%**, followed by **Random Forest** at **54% accuracy**. **Gradient Boosting** showed weaker performance at **46%**, while **Logistic Regression** performed the worst, with an accuracy of approximately **29%**, suggesting poor adaptability to the new dataset.

An important observation across both datasets is the consistent performance of the Support Vector Machine (SVM) model. While SVM did not achieve the highest accuracy in either the Kaggle test set or the external survey dataset, it demonstrated stable and competitive performance in both environments.

These findings highlight a clear contrast between **in-domain performance** (Kaggle dataset) and **out-of-domain performance** (survey dataset). While ensemble methods such as Random Forest and Gradient Boosting performed best on the Kaggle test set, their performance dropped significantly when applied to unseen, independently collected data.

Interestingly, **KNN emerged as the top-performing model on the survey dataset**, despite being one of the weaker models on Kaggle. This suggests that instance-based learning may be more robust in scenarios where the new data distribution differs from the training data.

Overall, the performance gap indicates that models trained on the Kaggle dataset did not generalize effectively to the survey dataset. This is likely due to differences in data distribution, sample size, feature characteristics, and potential noise in self-reported survey responses. The results emphasize the importance of testing machine learning models on external datasets before real-world deployment.

7. Conclusion

This study demonstrates that while machine learning models can effectively classify stress levels, their performance is highly dependent on the nature and quality of the underlying data. Across both datasets, behavioral patterns such as **higher screen time and lower sleep duration** consistently showed a strong association with elevated stress levels, reinforcing their relevance as key predictive features.

On the synthetic Kaggle dataset, all models achieved strong performance, with **Random Forest** emerging as the top performer. This indicates that nonlinear ensemble methods are highly effective at capturing structured patterns in large, well-defined datasets. However, when these same models were evaluated on an independently collected survey dataset, performance declined significantly across all approaches. This highlights a clear **generalization gap**, where models trained on synthetic or idealized data fail to fully translate to real-world scenarios.

Interestingly, while **KNN performed best on the survey dataset**, its overall accuracy remained modest. In contrast, **SVM demonstrated the most consistent performance across both datasets**, suggesting that models with balanced complexity may generalize more reliably under distributional shifts.

These findings underscore several important limitations. The Kaggle dataset, being artificially generated, may not accurately reflect real human behavior, while the survey dataset was relatively small and limited to a specific population. Additionally, the feature set captured only broad behavioral indicators, potentially missing more nuanced factors influencing stress.

Future work should focus on collecting **larger, more diverse, and real-world datasets**, incorporating **finer-grained behavioral features** (such as detailed digital usage patterns), and applying **feature selection or dimensionality reduction techniques** to improve model robustness.

Overall, this study highlights that while machine learning is a powerful tool for stress classification, **its real-world effectiveness depends on realistic data, thoughtful feature engineering, and careful model selection aligned with the data distribution**.

8. References

[1] American Institute of Stress. (2022, April 20). *What is the true cost of work-related stress?* https://www.stress.org/news/what-is-the-true-cost-of-work-related-stress/

[2] American Institute of Stress. (2022, April 20). *What is the true cost of work-related stress?* https://www.stress.org/news/what-is-the-true-cost-of-work-related-stress